Derivation Overview

Step 1: Variance of $Z(s+h) - Z(s)$

Key Formula (from probability theory)

For any two random variables X and Y:

$$\boxed{\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)}$$

Apply with $X = Z(s+h)$ and $Y = Z(s)$:

$$\mathrm{Var}\big(Z(s+h) - Z(s)\big) = \mathrm{Var}(Z(s+h)) + \mathrm{Var}(Z(s)) - 2\,\mathrm{Cov}\big(Z(s+h), Z(s)\big)$$

This is the **starting point**. No assumptions yet this is always true.

Step 2: Apply Second-Order Stationarity

Under **second-order stationarity** (weak stationarity):

1. **Constant variance:**

$$\mathrm{Var}(Z(s)) = \mathrm{Var}(Z(s+h)) = C(0) \quad \text{for all } s, h$$

   where $C(0)$ is the total variance (the covariance at lag 0).

2. **Covariance depends only on lag:**

$$\mathrm{Cov}\big(Z(s+h), Z(s)\big) = C(h) \quad \text{for all } s$$

   where $C(h)$ is the covariance function.

Step 3: Substitute into the Formula

Replace each term:

$$
\begin{aligned}
\mathrm{Var}\big(Z(s+h) - Z(s)\big) &= \underbrace{\mathrm{Var}(Z(s+h))}_{C(0)} + \underbrace{\mathrm{Var}(Z(s))}_{C(0)} - 2\,\underbrace{\mathrm{Cov}\big(Z(s+h), Z(s)\big)}_{C(h)} \\
&= C(0) + C(0) - 2C(h) \\
&= 2C(0) - 2C(h)
\end{aligned}
$$

Intermediate Result

$$
\boxed{\mathrm{Var}\big(Z(s+h) - Z(s)\big) = 2\big[C(0) - C(h)\big]}
$$

Step 4: Factor Out the 2

$$\mathrm{Var}\big(Z(s+h) - Z(s)\big) = 2\big[C(0) - C(h)\big]$$

Interpretation:

- The variance of the difference between two points is **twice** the difference between the total variance $C(0)$ and the covariance $C(h)$
- When $C(h)$ is large (points are correlated), the difference variance is small
- When $C(h) \approx 0$ (no correlation), the difference variance $\approx 2C(0)$

Now we decide to define a new function the **semivariogram**.

Step 5: Define the Semivariogram $\gamma(h)$

Definition (Matheron, 1962)

The **semivariogram** (usually just called variogram) is defined as:

$$\boxed{\gamma(h) = \frac{1}{2}\operatorname{Var}\big(Z(s+h) - Z(s)\big)}$$

Why the $\frac{1}{2}$?

- It removes the factor of 2 we saw in the previous step
- It makes $\gamma(h)$ equal the **sill** when $C(h) = 0$
- It gives a direct relationship with the covariance function

Step 6: Substitute the Variance Expression

We just found:

$$\mathrm{Var}\big(Z(s+h) - Z(s)\big) = 2\big[C(0) - C(h)\big]$$

Now plug into the definition of $\gamma(h)$:

$$\begin{aligned}
\gamma(h) &= \frac{1}{2} \times \mathrm{Var}\big(Z(s+h) - Z(s)\big) \\
&= \frac{1}{2} \times 2\big[C(0) - C(h)\big] \\
&= C(0) - C(h)
\end{aligned}$$

Key Result

$$\boxed{\gamma(h) = C(0) - C(h)}$$

Step 7: Interpret the Result

$$\gamma(h) = C(0) - C(h)$$

- **At $h = 0$:**

$$C(0) = \text{total variance}, \quad C(0) = \text{covariance at lag } 0$$

$$\gamma(0) = C(0) - C(0) = 0 \quad \checkmark$$

- **As $h \to \infty$ (no spatial correlation):**

$$C(h) \to 0 \quad \Rightarrow \quad \gamma(\infty) = C(0) - 0 = C(0)$$

This is the **sill** the total variance of the process.

Visual Summary of the Relationship

Lag h	$\boldsymbol{C(h)}$ **(Covariance)**	$\boldsymbol{\gamma(h)}$ **(Variogram)**
0	$C(0)$ (total variance)	0
Small	High (strong correlation)	Small
Medium	Moderate	Moderate
Large (Range)	Close to 0	Close to $C(0)$
∞	0	$C(0)$ (sill)

The variogram **behaves inversely** to the covariance function, shifted so that $\gamma(0) = 0$ and $\gamma(\infty) = C(0)$.

Summary: The Complete Derivation

1. Start with $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$
2. Apply to $Z(s + h)$ and $Z(s)$
3. Use second-order stationarity: $\text{Var} = C(0)$, $\text{Cov} = C(h)$
4. Obtain $\text{Var}(Z(s + h) - Z(s)) = 2[C(0) - C(h)]$
5. **Define** $\gamma(h) = \frac{1}{2}\text{Var}(Z(s + h) - Z(s))$
6. Get the fundamental relationship:

$$\boxed{\gamma(h) = C(0) - C(h)}$$

This holds **only** under second-order stationarity. For intrinsic processes, the variogram exists, but $C(h)$ may not.